

## Initiating Search

November 10, 2025, 8:51 PM

## • Search:

CAS Lexicon Search:

Sulfur fluoride exchange - Preferred Concept

## Search Tasks

| Task                                                         | Result Type                                                                                          | View                         |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------|
| Returned Reference Results from CAS Lexicon (212)            |  <b>References</b> | <a href="#">View Results</a> |
| Exported: Retrieved Related Reaction Results + Filters (296) |  <b>Reactions</b>  | <a href="#">View Results</a> |
| Filtered By:                                                 |                                                                                                      |                              |
| Yield:                                                       | 90-100%, 80-89%, 70-79%                                                                              |                              |
| Document Type:                                               | Journal                                                                                              |                              |

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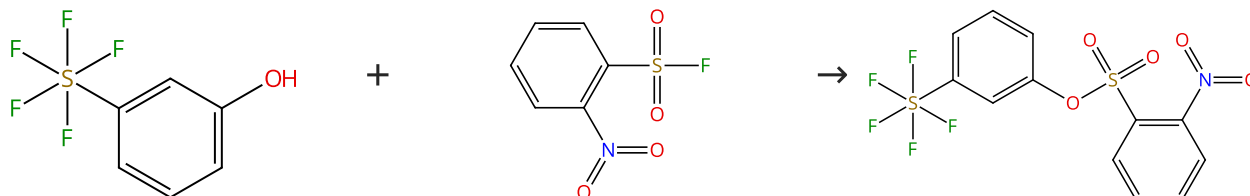
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# Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (46)

 Suppliers (54)

31-614-CAS-31115049

Steps: 1 Yield: 100%

Accelerated SuFEx Click Chemistry For Modular Synthesis

1.1 **Reagents:** Hexamethyldisilazane  
**Catalysts:** 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
**Solvents:** Acetonitrile; 5 min, rt

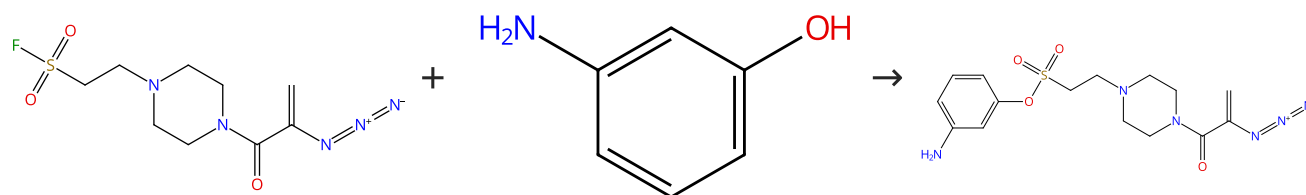
By: Smedley, Christopher J.; et al

Angewandte Chemie, International Edition (2022), 61(4),  
 e202112375.

[Experimental Protocols](#)

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (81)

31-614-CAS-40396715

Steps: 1 Yield: 100%

Iterative click reactions using trivalent platforms for sequential molecular assembly

1.1 **Catalysts:** 1,8-Diazabicyclo[5.4.0]undec-7-ene  
**Solvents:** Acetonitrile; 24 h, rt

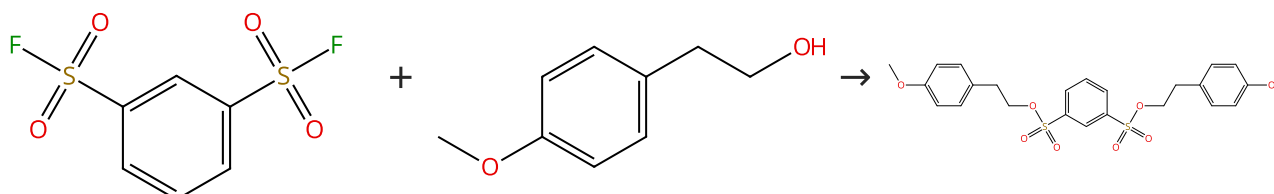
By: Orimoto, Gaku; et al

Chemical Communications (Cambridge, United Kingdom)  
 (2024), 60(45), 5824-5827.

[Experimental Protocols](#)

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

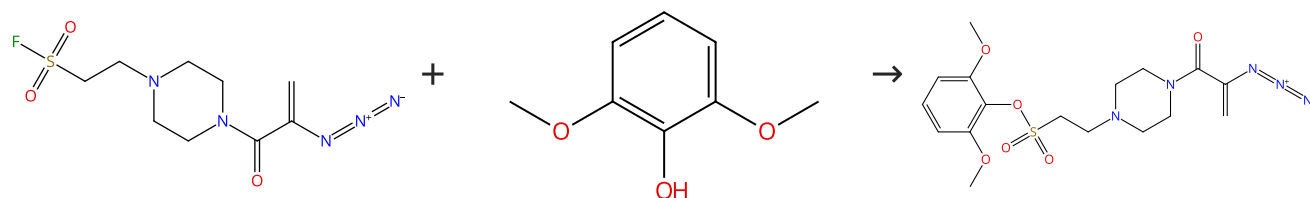

 Suppliers (34)

 Suppliers (86)

|                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-42835609</b><br>Steps: 1 Yield: 100%<br>1.1 <b>Reagents:</b> 1,8-Diazabicyclo[5.4.0]undec-7-ene<br><b>Solvents:</b> Acetonitrile- <i>d</i> <sub>3</sub> ; 5 min, rt<br>Experimental Protocols | <b>Benzene-1,3-disulfonyl fluoride and Benzene-1,3,5-trisulfonyl fluoride: Low-Cost, Stable, and Selective Reagents for SuFEx-Driven Deoxyazidation</b><br>By: Vishwakarma, Dharmendra S.; et al<br>Advanced Synthesis & Catalysis (2024), 366(21), 4470-4477. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%

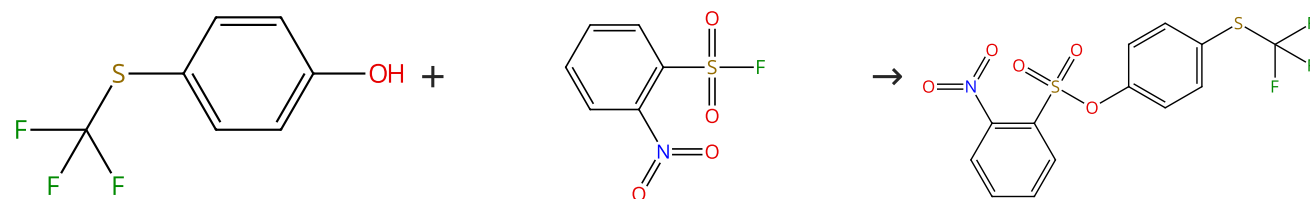


Suppliers (98)

|                                                                                                                                                                                     |                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-40396726</b><br>Steps: 1 Yield: 100%<br>1.1 <b>Catalysts:</b> 1,8-Diazabicyclo[5.4.0]undec-7-ene<br><b>Solvents:</b> Acetonitrile; 24 h, rt<br>Experimental Protocols | <b>Iterative click reactions using trivalent platforms for sequential molecular assembly</b><br>By: Orimoto, Gaku; et al<br>Chemical Communications (Cambridge, United Kingdom) (2024), 60(45), 5824-5827. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



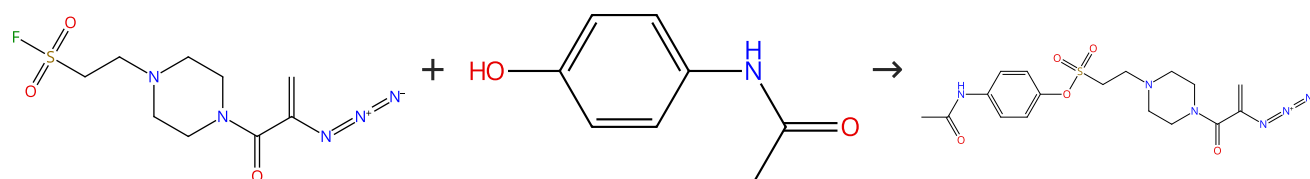
Suppliers (77)

Suppliers (54)

|                                                                                                                                                                                                                                               |                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-31115074</b><br>Steps: 1 Yield: 100%<br>1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 5 min, rt<br>Experimental Protocols | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b><br>By: Smedley, Christopher J.; et al<br>Angewandte Chemie, International Edition (2022), 61(4), e202112375. |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%

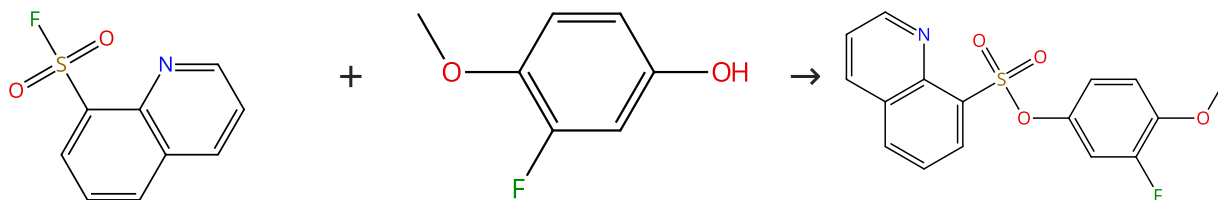


Suppliers (142)

|                                                                                                                                                                                     |                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-40396714</b><br>Steps: 1 Yield: 100%<br>1.1 <b>Catalysts:</b> 1,8-Diazabicyclo[5.4.0]undec-7-ene<br><b>Solvents:</b> Acetonitrile; 24 h, rt<br>Experimental Protocols | <b>Iterative click reactions using trivalent platforms for sequential molecular assembly</b><br>By: Orimoto, Gaku; et al<br>Chemical Communications (Cambridge, United Kingdom) (2024), 60(45), 5824-5827. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



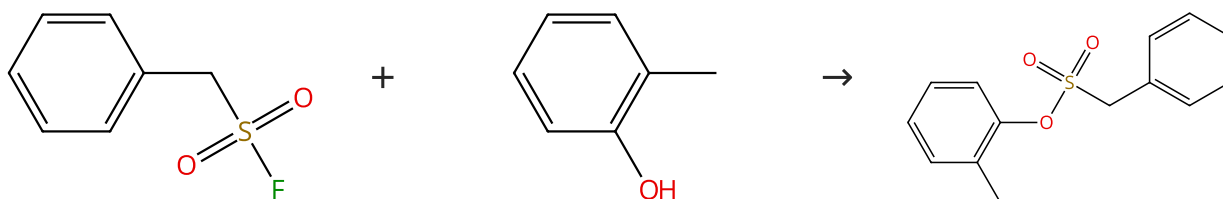
Suppliers (32)

Suppliers (60)

|                                                                                                                                                                                                                                                |                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-31115078</b><br>Steps: 1 Yield: 100%<br>1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 30 min, rt<br>Experimental Protocols | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b><br>By: Smedley, Christopher J.; et al<br>Angewandte Chemie, International Edition (2022), 61(4), e202112375. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (125)

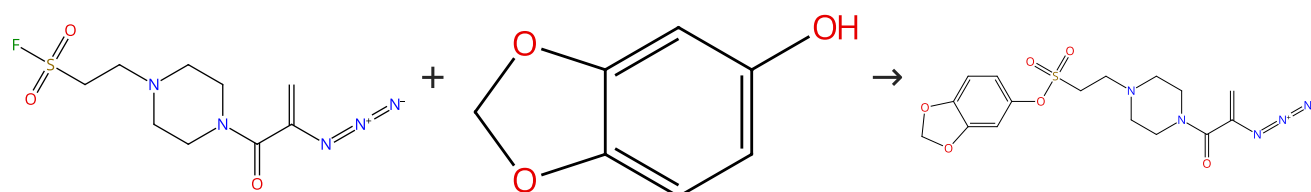
Suppliers (81)

Suppliers (3)

|                                                                                                                                                                                                                                                   |                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>31-614-CAS-31115106</b><br>Steps: 1 Yield: 100%<br>1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 30 min, 60 °C<br>Experimental Protocols | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b><br>By: Smedley, Christopher J.; et al<br>Angewandte Chemie, International Edition (2022), 61(4), e202112375. |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%

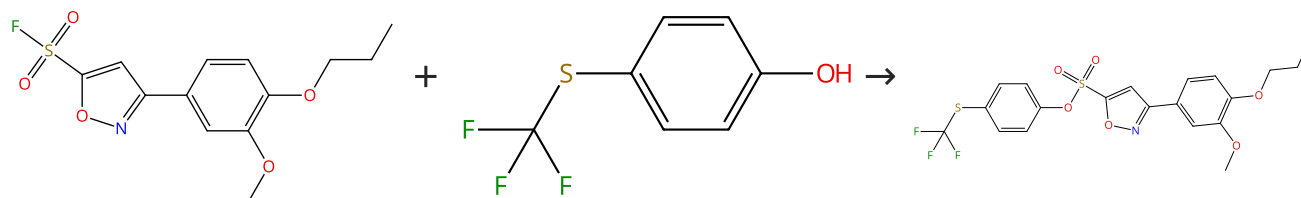


Suppliers (109)

|                                                                                       |                      |                                                                                       |
|---------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------|
| 31-614-CAS-40396711                                                                   | Steps: 1 Yield: 100% | Iterative click reactions using trivalent platforms for sequential molecular assembly |
| 1.1 Catalysts: 1,8-Diazabicyclo[5.4.0]undec-7-ene<br>Solvents: Acetonitrile; 24 h, rt |                      | By: Orimoto, Gaku; et al                                                              |
| Experimental Protocols                                                                |                      | Chemical Communications (Cambridge, United Kingdom) (2024), 60(45), 5824-5827.        |

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%

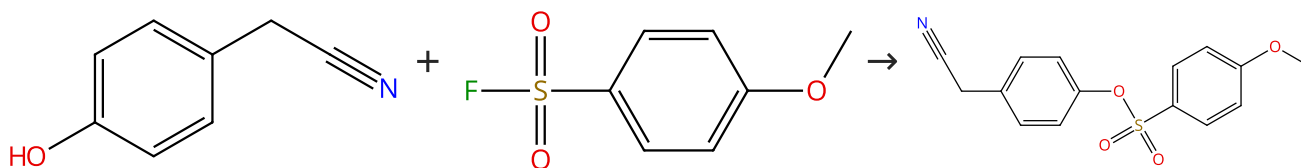


Suppliers (77)

|                                                                                                                                          |                      |                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------|
| 31-614-CAS-31115153                                                                                                                      | Steps: 1 Yield: 100% | Accelerated SuFEx Click Chemistry For Modular Synthesis             |
| 1.1 Reagents: Hexamethyldisilazane<br>Catalysts: 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br>Solvents: Acetonitrile; 5 min, rt |                      | By: Smedley, Christopher J.; et al                                  |
| Experimental Protocols                                                                                                                   |                      | Angewandte Chemie, International Edition (2022), 61(4), e202112375. |

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



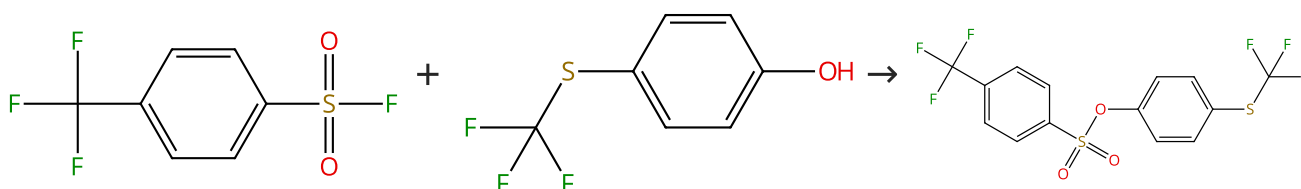
Suppliers (84)

Suppliers (33)

|                                                                                                                                           |                      |                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------|
| 31-614-CAS-31115067                                                                                                                       | Steps: 1 Yield: 100% | Accelerated SuFEx Click Chemistry For Modular Synthesis             |
| 1.1 Reagents: Hexamethyldisilazane<br>Catalysts: 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br>Solvents: Acetonitrile; 30 min, rt |                      | By: Smedley, Christopher J.; et al                                  |
| Experimental Protocols                                                                                                                    |                      | Angewandte Chemie, International Edition (2022), 61(4), e202112375. |

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



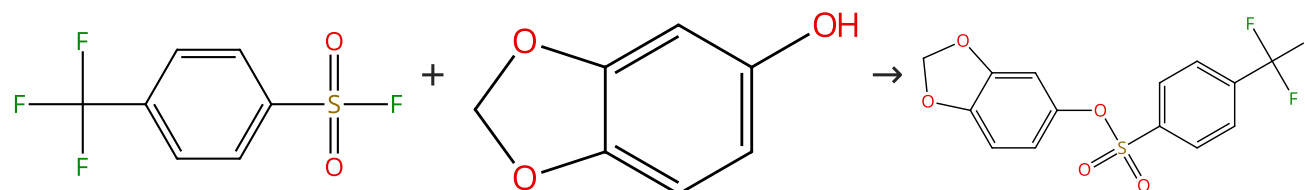
Suppliers (43)

Suppliers (77)

|                                                                                                                                                               |                      |                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------|
| 31-614-CAS-31115068                                                                                                                                           | Steps: 1 Yield: 100% | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b>                                            |
| 1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 5 min, rt |                      | By: Smedley, Christopher J.; et al<br>Angewandte Chemie, International Edition (2022), 61(4), e202112375. |
| Experimental Protocols                                                                                                                                        |                      |                                                                                                           |

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



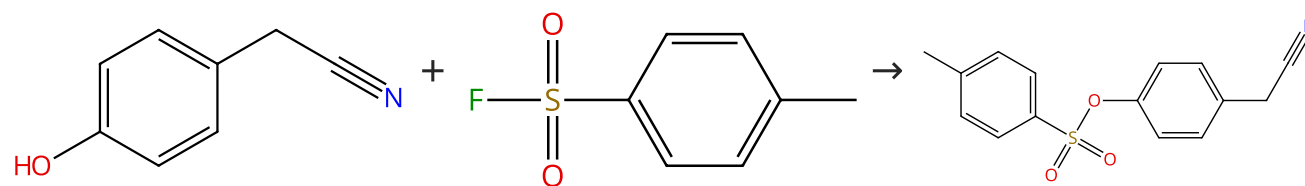
Suppliers (43)

Suppliers (109)

|                                                                                                                                                               |                      |                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------|
| 31-614-CAS-31115062                                                                                                                                           | Steps: 1 Yield: 100% | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b>                                            |
| 1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 5 min, rt |                      | By: Smedley, Christopher J.; et al<br>Angewandte Chemie, International Edition (2022), 61(4), e202112375. |
| Experimental Protocols                                                                                                                                        |                      |                                                                                                           |

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

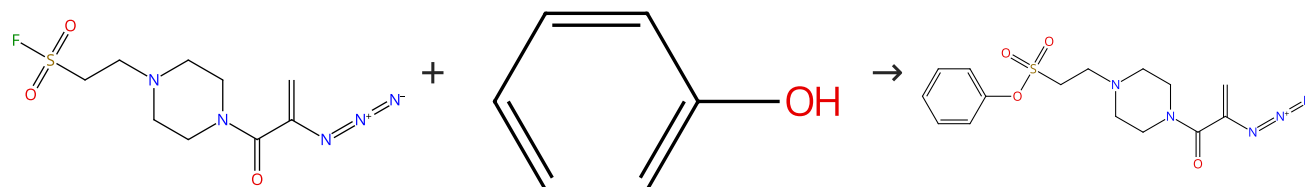
Suppliers (69)

Suppliers (21)

|                                                                                                                                                               |                      |                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------|
| 31-614-CAS-31115081                                                                                                                                           | Steps: 1 Yield: 100% | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b>                                            |
| 1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 5 min, rt |                      | By: Smedley, Christopher J.; et al<br>Angewandte Chemie, International Edition (2022), 61(4), e202112375. |
| Experimental Protocols                                                                                                                                        |                      |                                                                                                           |

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%

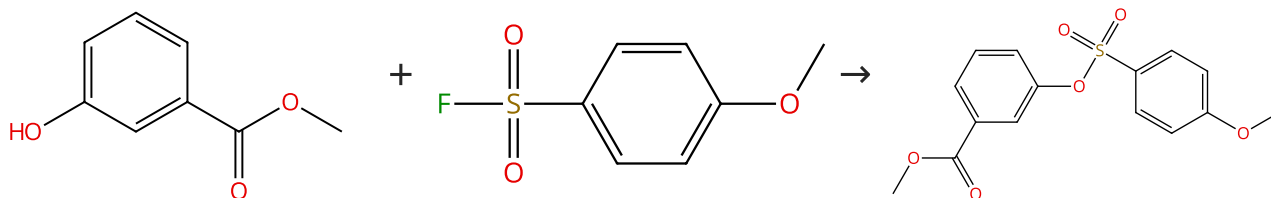


Suppliers (207)

|                                                                                                     |                      |                                                                                              |
|-----------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------|
| 31-614-CAS-40396712                                                                                 | Steps: 1 Yield: 100% | <b>Iterative click reactions using trivalent platforms for sequential molecular assembly</b> |
| 1.1 <b>Catalysts:</b> 1,8-Diazabicyclo[5.4.0]undec-7-ene<br><b>Solvents:</b> Acetonitrile; 24 h, rt |                      | By: Orimoto, Gaku; et al                                                                     |
| Experimental Protocols                                                                              |                      | Chemical Communications (Cambridge, United Kingdom) (2024), 60(45), 5824-5827.               |

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



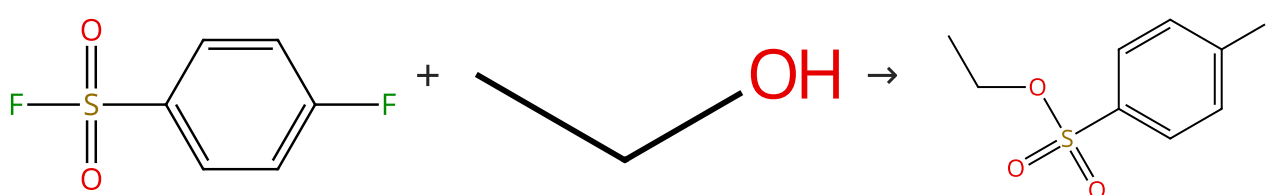
Suppliers (90)

Suppliers (33)

|                                                                                                                                                                |                      |                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------|
| 31-614-CAS-31115073                                                                                                                                            | Steps: 1 Yield: 100% | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b>      |
| 1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 60 min, rt |                      | By: Smedley, Christopher J.; et al                                  |
| Experimental Protocols                                                                                                                                         |                      | Angewandte Chemie, International Edition (2022), 61(4), e202112375. |

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (44)

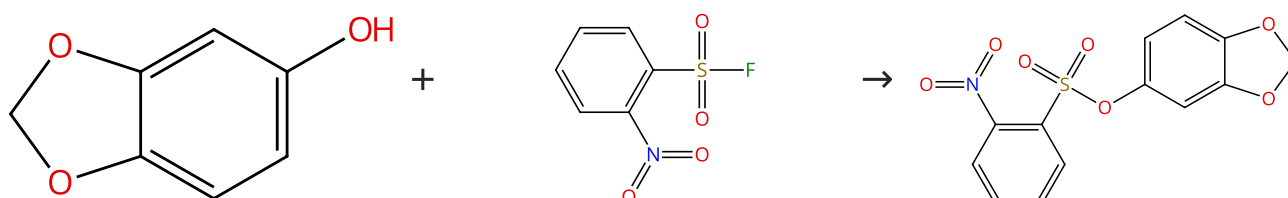
Suppliers (620)

Suppliers (7)

|                                                                                                                                                                |                      |                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------|
| 31-614-CAS-31115094                                                                                                                                            | Steps: 1 Yield: 100% | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b>      |
| 1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 30 min, rt |                      | By: Smedley, Christopher J.; et al                                  |
| Experimental Protocols                                                                                                                                         |                      | Angewandte Chemie, International Edition (2022), 61(4), e202112375. |

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



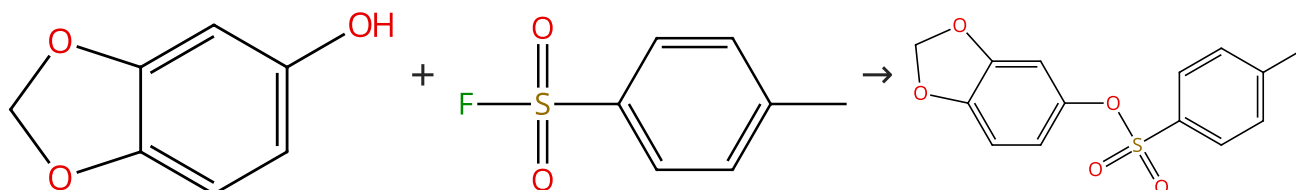
Suppliers (109)

Suppliers (54)

|                                                                                                                                                               |                      |                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------|
| 31-614-CAS-31115071                                                                                                                                           | Steps: 1 Yield: 100% | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b>                                            |
| 1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 5 min, rt |                      | By: Smedley, Christopher J.; et al<br>Angewandte Chemie, International Edition (2022), 61(4), e202112375. |
| Experimental Protocols                                                                                                                                        |                      |                                                                                                           |

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (109)

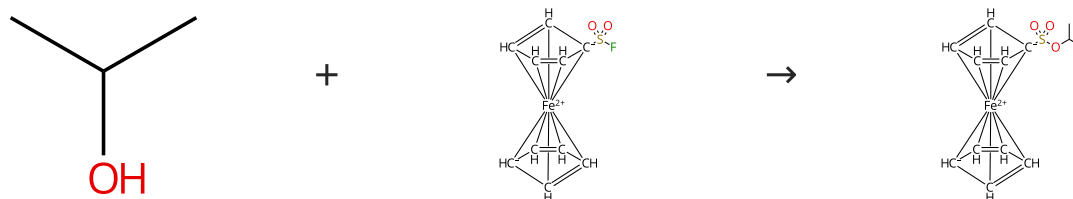
Suppliers (69)

Suppliers (11)

|                                                                                                                                                                |                      |                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------|
| 31-614-CAS-31115066                                                                                                                                            | Steps: 1 Yield: 100% | <b>Accelerated SuFEx Click Chemistry For Modular Synthesis</b>                                            |
| 1.1 <b>Reagents:</b> Hexamethyldisilazane<br><b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 10 min, rt |                      | By: Smedley, Christopher J.; et al<br>Angewandte Chemie, International Edition (2022), 61(4), e202112375. |
| Experimental Protocols                                                                                                                                         |                      |                                                                                                           |

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%

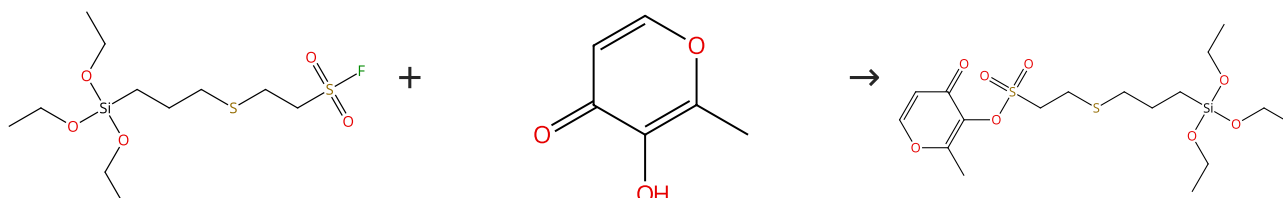


Suppliers (365)

|                                                                                               |                     |                                                                             |
|-----------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------|
| 31-614-CAS-31086910                                                                           | Steps: 1 Yield: 99% | <b>The chemistry of ferrocenesulfonyl fluoride revealed</b>                 |
| 1.1 <b>Reagents:</b> Sodium hydride<br><b>Solvents:</b> Tetrahydrofuran; 25 °C; 15 min, 25 °C |                     | By: Erb, William; et al<br>Dalton Transactions (2021), 50(45), 16483-16487. |
| 1.2 30 min, 25 °C                                                                             |                     |                                                                             |
| Experimental Protocols                                                                        |                     |                                                                             |

Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



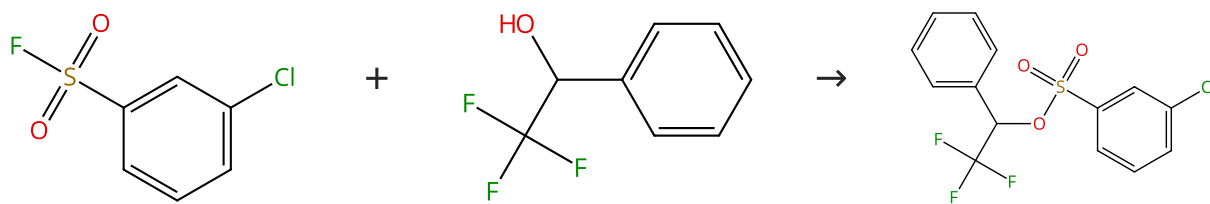
Suppliers (105)

|                                                                                                                    |                     |                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------|
| 31-614-CAS-41545957                                                                                                | Steps: 1 Yield: 99% | <b>Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking</b> |
| 1.1 <b>Catalysts:</b> 2- <i>tert</i> -Butyl-1,1,3,3-tetramethylguanidine<br><b>Solvents:</b> Acetonitrile; 2 h, rt |                     | By: Wang, Minlong; et al<br>Nature Communications (2024), 15(1), 6849.                                          |
| Experimental Protocols                                                                                             |                     |                                                                                                                 |



Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (33)

Suppliers (80)

31-614-CAS-44586002

Steps: 1 Yield: 99%

**N-Heterocyclic carbene-catalyzed SuFEx reactions of fluoroalkylated secondary benzylic alcohols**1.1 **Catalysts:** 1,3-Dihydro-1,3-bis(1-methylethyl)-2*H*-imidazol-2-ylidene**Solvents:** Acetonitrile; 24 h, rt

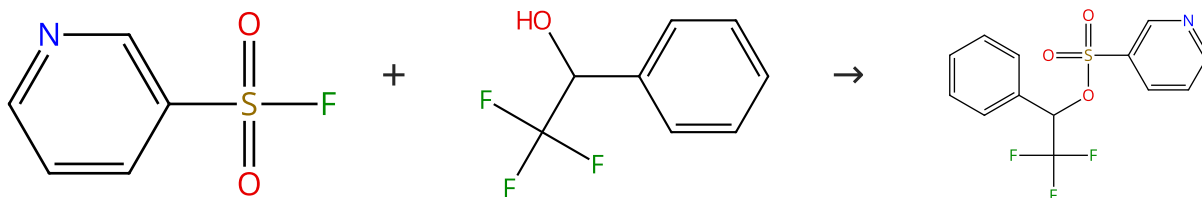
By: Wei, Zhihang; et al

Organic &amp; Biomolecular Chemistry (2025), 23(14), 3465-3469.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (34)

Suppliers (80)

31-614-CAS-44586009

Steps: 1 Yield: 99%

**N-Heterocyclic carbene-catalyzed SuFEx reactions of fluoroalkylated secondary benzylic alcohols**1.1 **Catalysts:** 1,3-Dihydro-1,3-bis(1-methylethyl)-2*H*-imidazol-2-ylidene**Solvents:** Acetonitrile; 24 h, rt

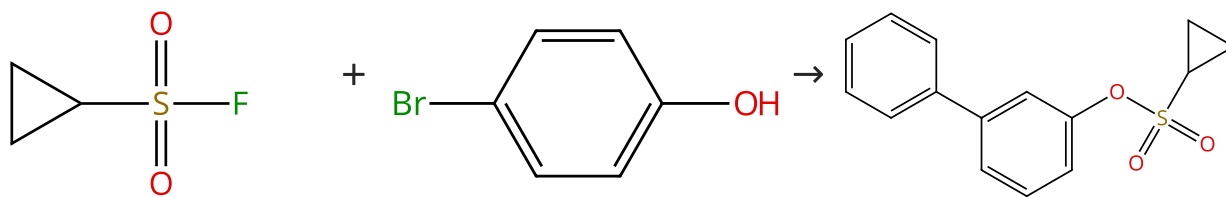
By: Wei, Zhihang; et al

Organic &amp; Biomolecular Chemistry (2025), 23(14), 3465-3469.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (25)

Suppliers (82)

Suppliers (3)

31-614-CAS-31115110

Steps: 1 Yield: 99%

**Accelerated SuFEx Click Chemistry For Modular Synthesis**1.1 **Reagents:** Hexamethyldisilazane**Catalysts:** 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine**Solvents:** Acetonitrile; 30 min, 60 °C

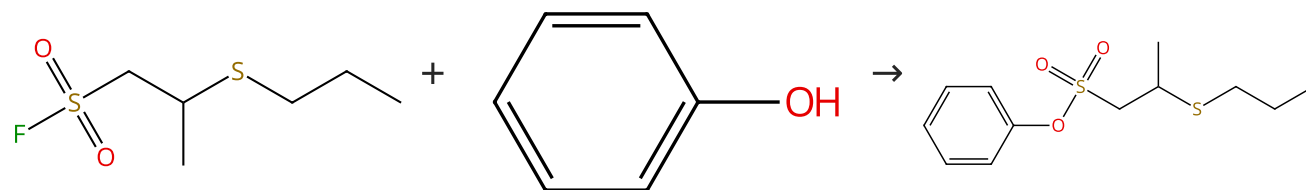
By: Smedley, Christopher J.; et al

Angewandte Chemie, International Edition (2022), 61(4), e202112375.

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (207)

Supplier (1)

31-614-CAS-41545972

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

By: Wang, Minlong; et al

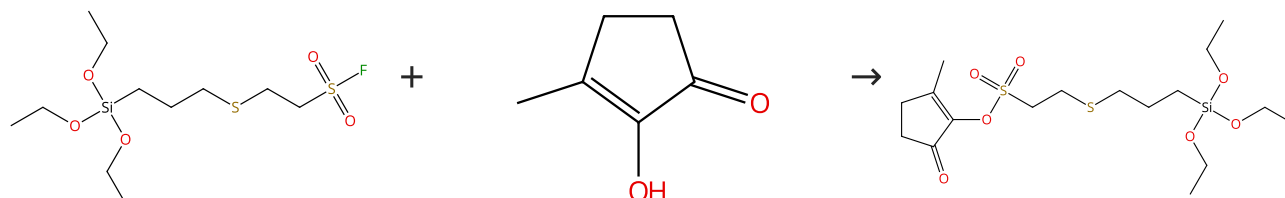
Nature Communications (2024), 15(1), 6849.

 1.1 **Reagents:** 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
**Solvents:** Acetonitrile; 0.5 h, rt

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (67)

31-614-CAS-41545950

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

By: Wang, Minlong; et al

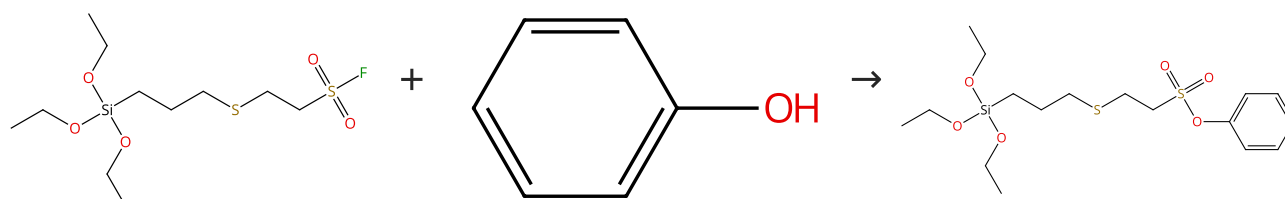
Nature Communications (2024), 15(1), 6849.

 1.1 **Catalysts:** 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
**Solvents:** Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (207)

31-614-CAS-41545948

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

By: Wang, Minlong; et al

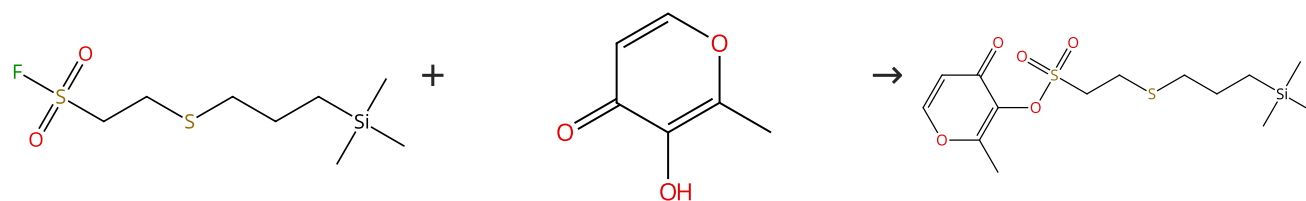
Nature Communications (2024), 15(1), 6849.

 1.1 **Catalysts:** 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
**Solvents:** Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (105)

31-614-CAS-41545912

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated Su FEx click chemistry for efficient organic-inorganic linking**

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

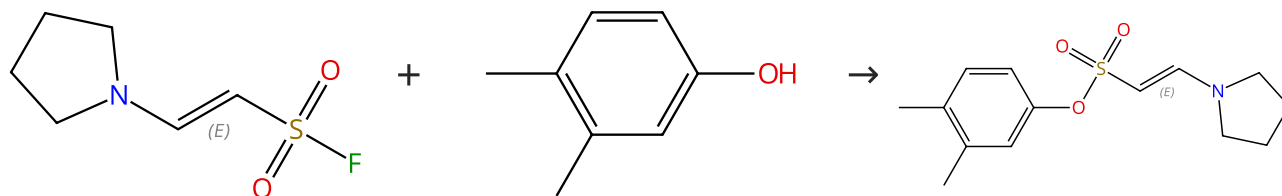
1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine

Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (69)

Double bond geometry shown

31-022-CAS-21992351

Steps: 1 Yield: 99%

**A Simple Protocol for the Stereoselective Construction of Enaminy Sulfonamide Fluorides**

By: Leng, Jing; et al

Organic Letters (2020), 22(11), 4316-4321.

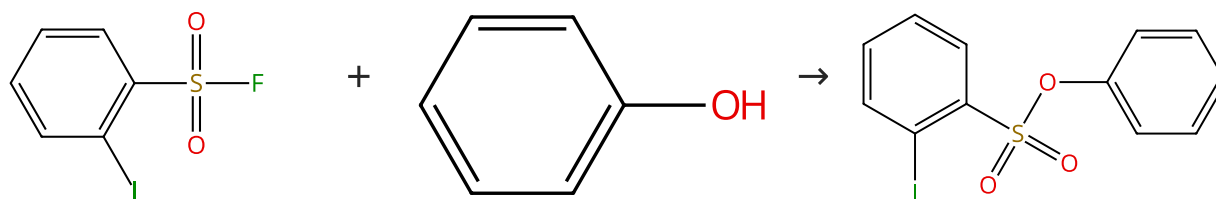
1.1 Reagents: Potassium hydroxide

Solvents: Acetonitrile; 5 - 15 h, 50 °C

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (17)

Suppliers (207)

Suppliers (3)

31-614-CAS-41545969

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated Su FEx click chemistry for efficient organic-inorganic linking**

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

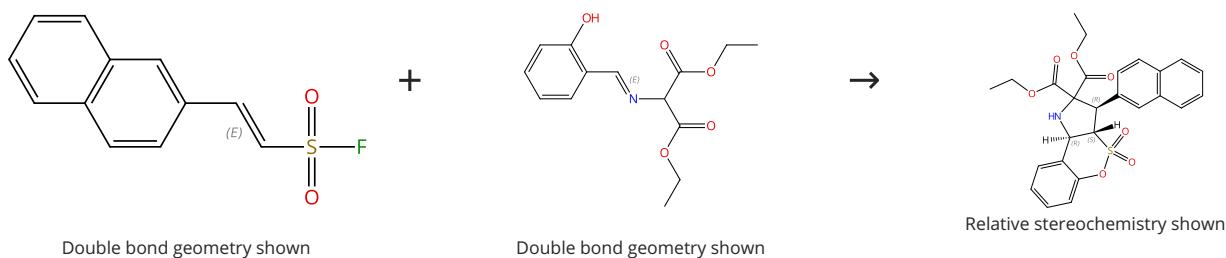
1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine

Solvents: Acetonitrile; 0.5 h, rt

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

31-614-CAS-41942649

Steps: 1 Yield: 99%

**Bronsted base-catalyzed assembly of sulfochromeno[4,3-b]pyrrolidines via tandem [3 + 2] cycloaddition-SuFEx click reaction of ethenesulfonyl fluorides and o-hydroxyaryl azomethines**

1.1 **Catalysts:** Triethylamine  
**Solvents:** Acetonitrile; 24 h, rt

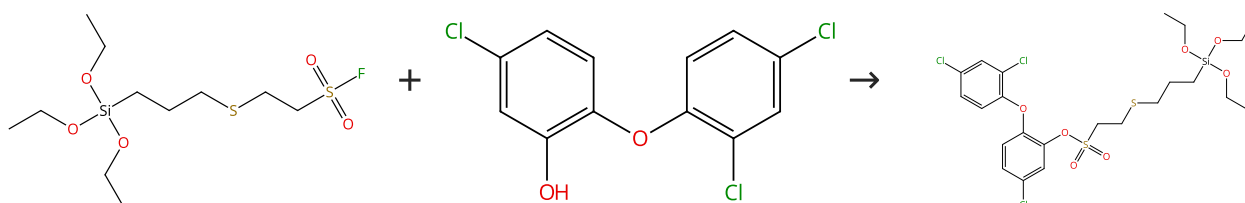
Experimental Protocols

By: Zhang, Fang; et al

Organic Chemistry Frontiers (2024), 11(21), 6177-6183.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (103)

31-614-CAS-41545956

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

1.1 **Catalysts:** 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
**Solvents:** Acetonitrile; 2 h, rt

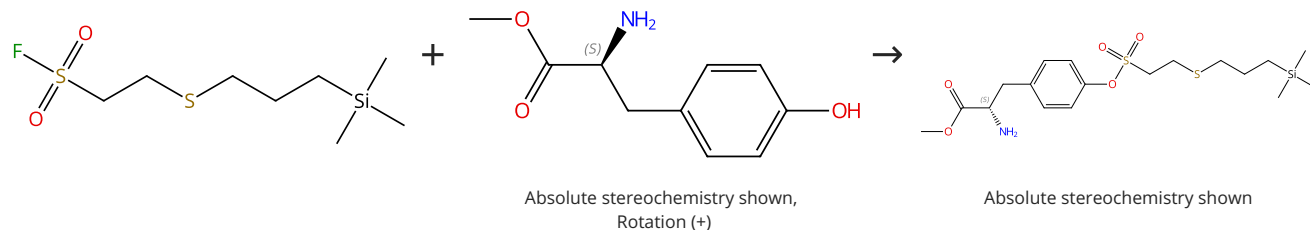
Experimental Protocols

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

31-614-CAS-41545890

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

1.1 **Reagents:** 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
**Solvents:** Acetonitrile; 2 h, rt

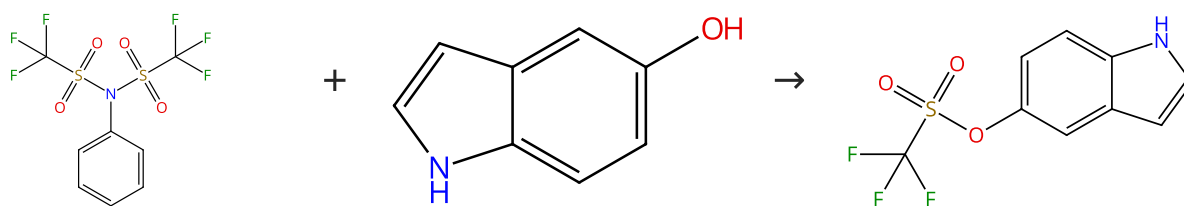
Experimental Protocols

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

## Scheme 34 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (98)

Suppliers (17)

31-614-CAS-31112521

Steps: 1 Yield: 99%

SuFEx-enabled, chemoselective synthesis of triflates, triflamides and triflimidates

By: Li, Bing-Yu; et al

Chemical Science (2022), 13(8), 2270-2279.

1.1 Reagents: Diisopropylethylamine, Potassium bifluoride

Solvents: Acetonitrile, Water; 18 h, rt

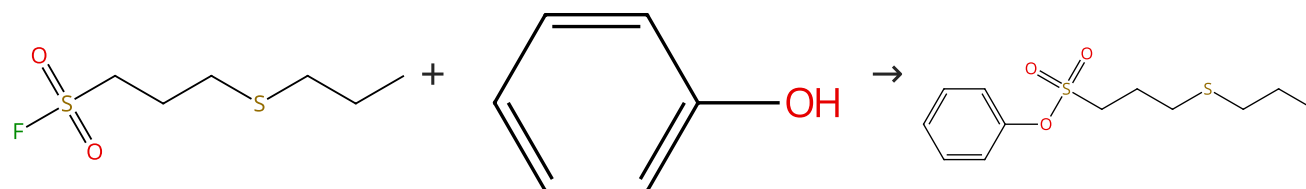
1.2 Reagents: Sodium hydroxide

Solvents: Water

Experimental Protocols

## Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (207)

31-614-CAS-41545980

Steps: 1 Yield: 99%

Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

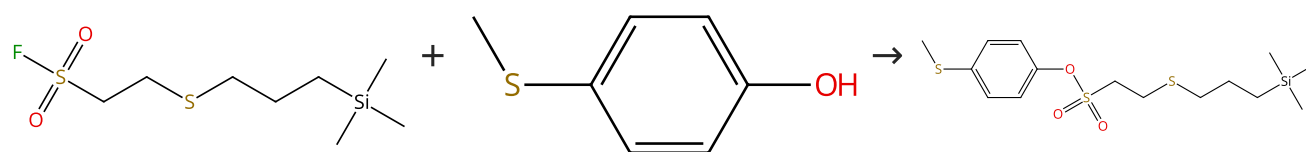
1.1 Reagents: 2-tert-Butyl-1,1,3,3-tetramethylguanidine

Solvents: Acetonitrile; 0.5 h, rt

Experimental Protocols

## Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (61)

31-614-CAS-41545874

Steps: 1 Yield: 99%

Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

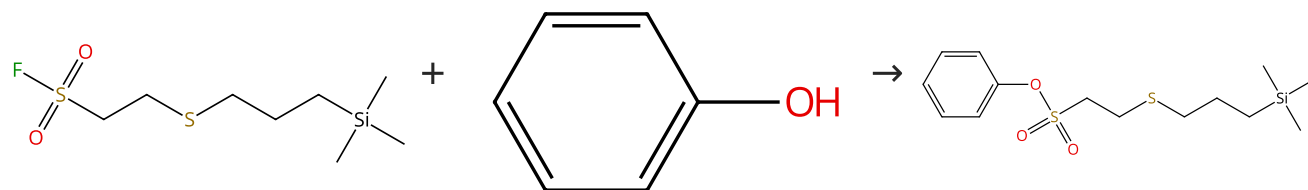
1.1 Reagents: 2-tert-Butyl-1,1,3,3-tetramethylguanidine

Solvents: Acetonitrile; 0.5 h, rt

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (207)

31-614-CAS-41545865

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

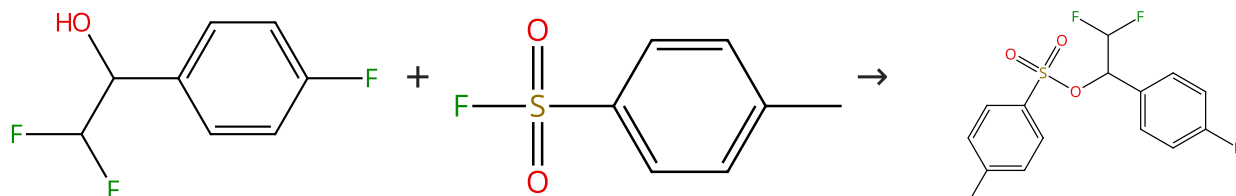
1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine

Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (18)

Suppliers (69)

31-614-CAS-44586016

Steps: 1 Yield: 99%

**N-Heterocyclic carbene-catalyzed SuFEx reactions of fluoroalkylated secondary benzylic alcohols**

By: Wei, Zhihang; et al

Organic &amp; Biomolecular Chemistry (2025), 23(14), 3465-3469.

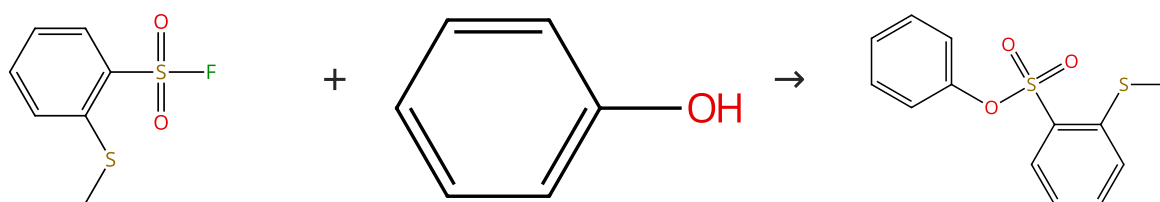
1.1 Catalysts: 1,3-Dihydro-1,3-bis(1-methylethyl)-2*H*-imidazol-2-ylidene

Solvents: Acetonitrile; 24 h, rt

Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (4)

Suppliers (207)

31-614-CAS-41545978

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

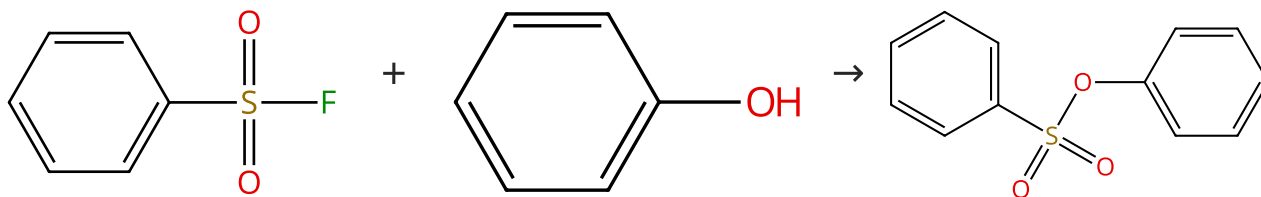
1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine

Solvents: Acetonitrile; 0.5 h, rt

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (46)

Suppliers (207)

Suppliers (31)

31-614-CAS-41545968

Steps: 1 Yield: 99%

Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking

1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
Solvents: Acetonitrile; 0.5 h, rt

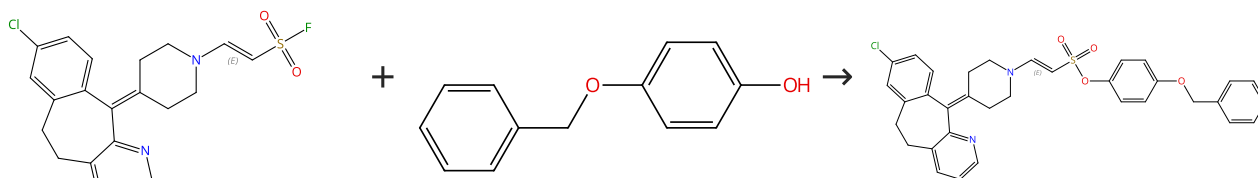
By: Wang, Minlong; et al

Experimental Protocols

Nature Communications (2024), 15(1), 6849.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (103)

Double bond geometry shown

31-022-CAS-21992354

Steps: 1 Yield: 99%

A Simple Protocol for the Stereoselective Construction of Enaminy Sulfonate Fluorides

1.1 Reagents: Potassium hydroxide  
Solvents: Acetonitrile; 5 - 15 h, 50 °C

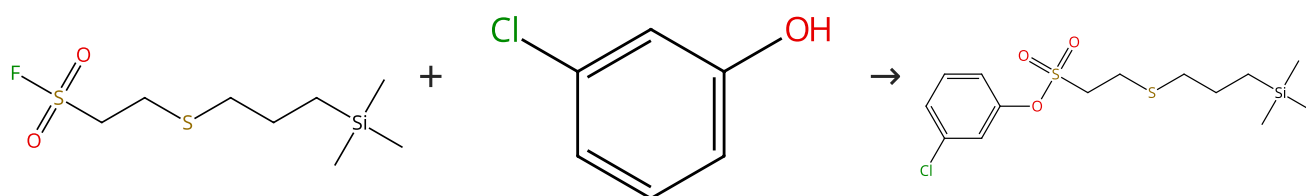
By: Leng, Jing; et al

Experimental Protocols

Organic Letters (2020), 22(11), 4316-4321.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

31-614-CAS-41545882

Steps: 1 Yield: 99%

Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking

1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
Solvents: Acetonitrile; 0.5 h, rt

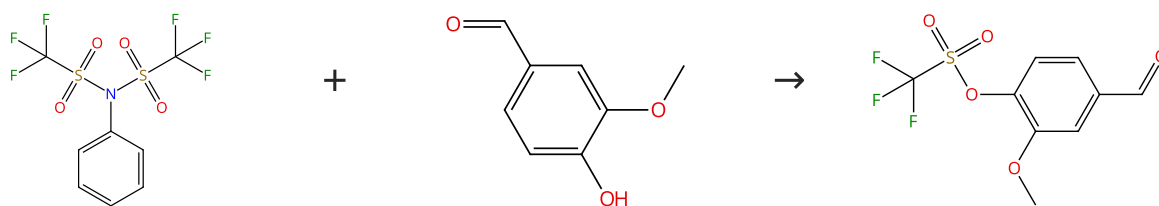
By: Wang, Minlong; et al

Experimental Protocols

Nature Communications (2024), 15(1), 6849.

## Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (145)

Suppliers (9)

31-614-CAS-31112519

Steps: 1 Yield: 99%

SuFEx-enabled, chemoselective synthesis of triflates, triflamides and triflimidates

By: Li, Bing-Yu; et al

Chemical Science (2022), 13(8), 2270-2279.

1.1 Reagents: Diisopropylethylamine, Potassium bifluoride

Solvents: Acetonitrile, Water; 12 h, rt

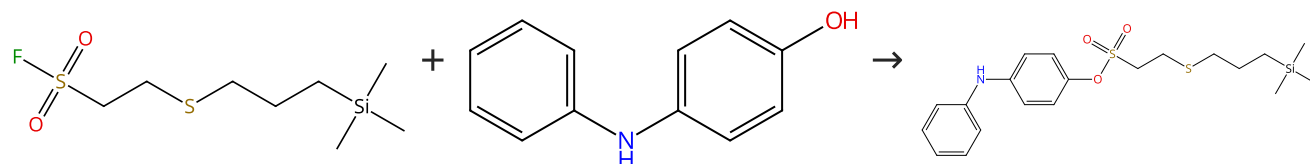
1.2 Reagents: Sodium hydroxide

Solvents: Water

Experimental Protocols

## Scheme 44 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (81)

31-614-CAS-41545891

Steps: 1 Yield: 99%

Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking

By: Wang, Minlong; et al

Nature Communications (2024), 15(1), 6849.

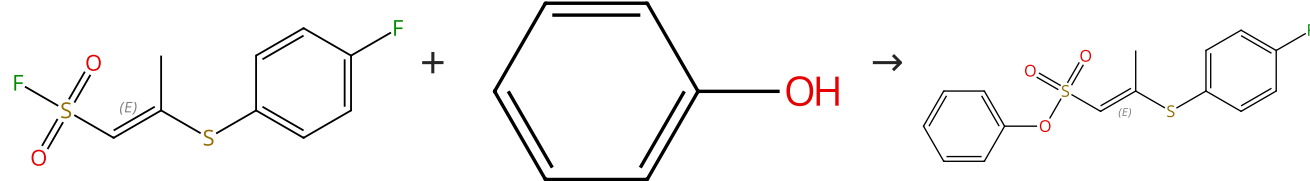
1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine

Solvents: Acetonitrile; 2 h, rt

Experimental Protocols

## Scheme 45 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (207)

Double bond geometry shown

31-614-CAS-44678683

Steps: 1 Yield: 99%

A simple protocol for stereoselective construction of novel  $\beta$ -sulfanyl vinyl sulfonate fluorides

By: Ajisafe, Monday Peter; et al

Chemical Communications (Cambridge, United Kingdom) (2025), 61(31), 5798-5801.

1.1 Reagents: Cesium carbonate

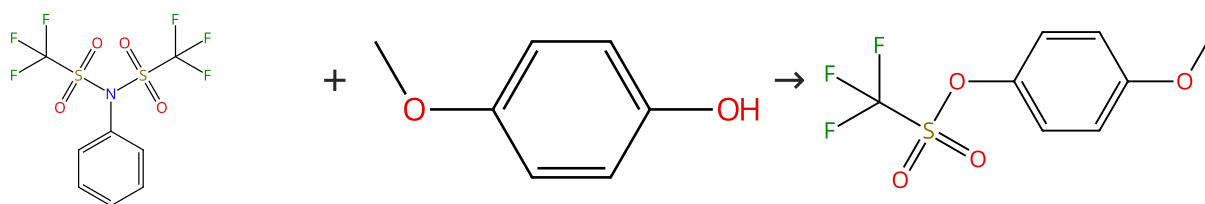
Solvents: Acetonitrile; 1 h, rt

Experimental Protocols



## Scheme 46 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

Suppliers (120)

Suppliers (59)

31-614-CAS-31112511

Steps: 1 Yield: 99%

SuFEx-enabled, chemoselective synthesis of triflates, triflamides and triflimidates

By: Li, Bing-Yu; et al

Chemical Science (2022), 13(8), 2270-2279.

1.1 Reagents: Diisopropylethylamine, Potassium bifluoride

Solvents: Acetonitrile, Water; 18 h, rt

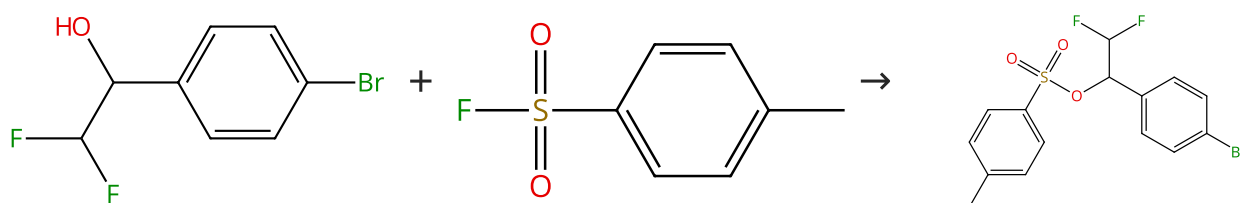
1.2 Reagents: Sodium hydroxide

Solvents: Water

Experimental Protocols

## Scheme 47 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (21)

Suppliers (69)

31-614-CAS-44586026

Steps: 1 Yield: 99%

N-Heterocyclic carbene-catalyzed SuFEx reactions of fluoroalkylated secondary benzylic alcohols

By: Wei, Zhihang; et al

Organic &amp; Biomolecular Chemistry (2025), 23(14), 3465-3469.

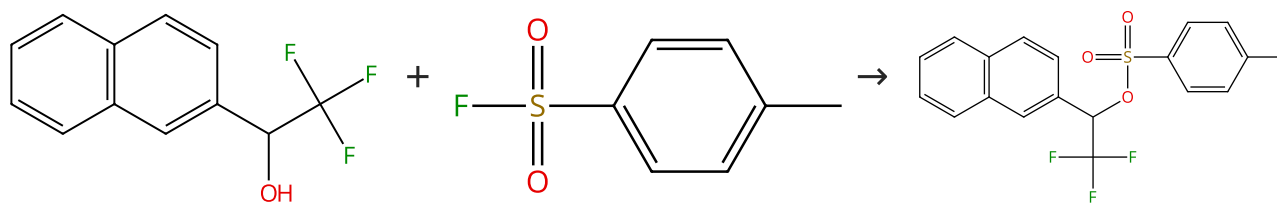
1.1 Catalysts: 1,3-Dihydro-1,3-bis(1-methylethyl)-2H-imidazol-2-ylidene

Solvents: Acetonitrile; 24 h, rt

Experimental Protocols

## Scheme 48 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (47)

Suppliers (69)

Suppliers (2)

31-614-CAS-44585993

Steps: 1 Yield: 99%

N-Heterocyclic carbene-catalyzed SuFEx reactions of fluoroalkylated secondary benzylic alcohols

By: Wei, Zhihang; et al

Organic &amp; Biomolecular Chemistry (2025), 23(14), 3465-3469.

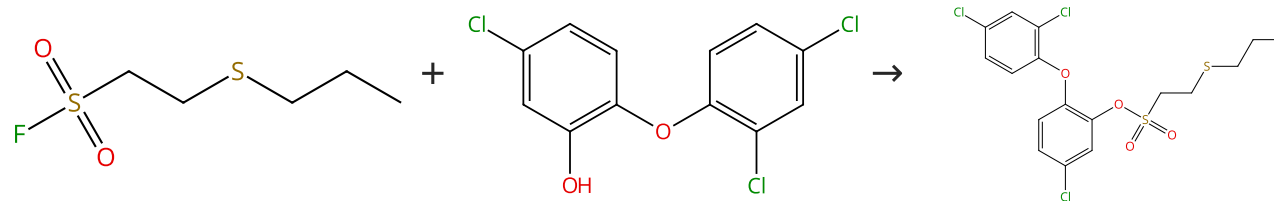
1.1 Catalysts: 1,3-Dihydro-1,3-bis(1-methylethyl)-2H-imidazol-2-ylidene

Solvents: Acetonitrile; 24 h, rt

Experimental Protocols

Scheme 49 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (103)

31-614-CAS-41545985

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
Solvents: Acetonitrile; 2 h, rt

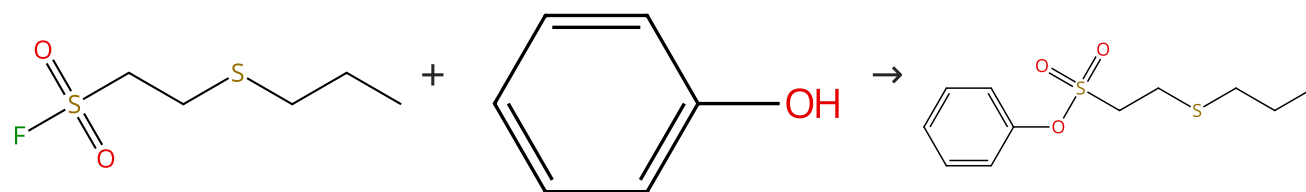
By: Wang, Minlong; et al

Experimental Protocols

Nature Communications (2024), 15(1), 6849.

Scheme 50 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (207)

31-614-CAS-41545870

Steps: 1 Yield: 99%

**Intramolecular chalcogen bonding activated SuFEx click chemistry for efficient organic-inorganic linking**

1.1 Reagents: 2-*tert*-Butyl-1,1,3,3-tetramethylguanidine  
Solvents: Acetonitrile; 2 h, rt

By: Wang, Minlong; et al

Experimental Protocols

Nature Communications (2024), 15(1), 6849.